FashionFlow AI

AI-Powered Fashion Commerce & Growth Platform

PRODUCT REQUIREMENTS DOCUMENT

Version 1.0 — Engineering Build Reference

Prepared for: Engineering, Design, QA, and Product Teams
Status: Single Source of Truth

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Product Requirements Document (PRD)

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1. Executive Summary

1.1 Overview

FashionFlow AI is a multi-tenant SaaS platform purpose-built for clothing stores, fashion brands, boutique owners, and online fashion businesses. It unifies product management, inventory, point-of-sale, order management, customer relationship management, marketing automation, analytics, and generative AI into a single operating system for fashion commerce.

Unlike general-purpose e-commerce platforms, FashionFlow AI is opinionated for the fashion vertical: every entity (product, variant, campaign, customer segment) is modeled around the realities of apparel retail — size/color matrices, seasonal collections, style-driven merchandising, and visually-led marketing.

1.2 The Business Problem

Fashion retailers today are forced to stitch together 4–6 disconnected tools to run their business:

Function	Typical Tool Used Today	Problem
Inventory & POS	Shopify, Square, Lightspeed	No fashion-specific variant/size intelligence
Marketing	Meta Ads Manager, Mailchimp	Not connected to real inventory or sales data
Design/Content	Canva, Photoshop, freelancers	Slow, expensive, disconnected from catalog
Customer Management	Spreadsheets, basic CRM	No segmentation tied to purchase behavior
Analytics	Google Analytics, manual spreadsheets	Fragmented, not fashion-specific, no AI insight

The result: store owners spend more time reconciling data across tools than growing their business, and they lack the marketing/design resources of larger brands.

1.3 Market Opportunity

The global apparel and fashion e-commerce market is large and growing, driven by:

- Increasing digitization of independent and multi-branch fashion retailers.
- Rising cost of hiring dedicated designers, marketers, and analysts for small/medium fashion businesses.
- Growing demand for AI-generated content (product copy, ad creative, design concepts) as a substitute for expensive creative resources.
- A gap in the market for a **vertical SaaS** platform (fashion-specific) versus horizontal platforms (Shopify, generic POS systems) that treat apparel like any other retail category.

1.4 Vision

FashionFlow AI's vision is to become the **default operating system for fashion businesses** — the single platform a boutique owner or fashion brand opens every morning to manage their store, understand their customers, run their marketing, and generate the creative content they need to compete with much larger brands.

1.5 Objectives

Objective	Description
O1 — Unify Operations	Replace 4–6 fragmented tools with one platform covering catalog, inventory, POS, CRM, and marketing.
O2 — Automate Growth	Use AI to automate content creation, campaign generation, and business recommendations.
O3 — Improve Decision-Making	Provide fashion-specific analytics and AI insights that a small business owner could not otherwise access.
O4 — Increase Retention & Revenue	Give merchants tools (segmentation, campaigns, personalized recommendations) that directly move sales and repeat-purchase rate.
O5 — Scale to Multi-Branch Retailers	Support multi-store, multi-branch operations under a single account with role-based access.

1.6 Success Metrics & KPIs

Category	KPI	Target Definition
Adoption	Active stores (weekly active)	Stores completing ≥1 core action (order, product edit, campaign) per week
Activation	Time-to-first-sale	Median time from signup to first recorded order
AI Engagement	AI feature adoption rate	% of active stores using ≥1 AI feature per month
Retention	Store churn rate	% of stores canceling subscription per month
Revenue Impact	Merchant GMV growth	% increase in merchant revenue quarter-over-quarter after adoption
Marketing Effectiveness	Campaign-attributed revenue	% of merchant revenue traceable to in-platform campaigns
Platform Health	System uptime	≥99.9% monthly uptime (see Non-Functional Requirements)

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2. Product Vision

2.1 Why FashionFlow AI Exists

Fashion is a visually driven, trend-sensitive, and inventory-complex retail category. Generic commerce platforms treat a t-shirt the same way they treat a bag of rice: as a SKU with a price and a quantity. This ignores the realities fashion merchants live with daily:

- A single style can explode into 20+ sellable variants (size × color × sometimes fit).
- Trends and seasons make certain products time-sensitive and perishable in demand.
- Visual merchandising and content (photos, descriptions, campaign creative) drive conversion far more than in other retail categories.
- Fashion customers respond strongly to segmentation (VIP, frequent buyer, seasonal shopper) and personalized offers.

FashionFlow AI exists to encode this domain knowledge directly into the product, rather than forcing merchants to configure a generic platform to approximate it.

2.2 How FashionFlow AI Differs

2.2.1 vs. Shopify

Dimension	Shopify	FashionFlow AI
Product Model	Generic product/variant model	Native size/color/fit variant matrix with fashion-specific attributes
Marketing	Requires third-party apps for campaigns	Native campaign engine tied directly to inventory and customer segments
AI Content	Limited/plugin-based	Native AI Product Writer, Fashion Designer, Campaign Generator
Analytics	Generic commerce analytics	Fashion-specific analytics (slow-moving styles, size sell-through, seasonal performance)
Design Tools	None (requires Canva/Photoshop)	Native AI Fashion Designer for prints, patterns, and collections

2.2.2 vs. Traditional POS Systems

Traditional POS systems (e.g., Lightspeed, Square) focus narrowly on the point of transaction: ringing up a sale, tracking stock at a register. They are not built to run marketing campaigns, manage customer relationships beyond a purchase log, or generate content. FashionFlow AI treats POS as one module among many, feeding the same real-time inventory, customer, and analytics engine used by marketing and AI.

2.2.3 vs. Standalone Inventory Systems

Standalone inventory systems solve stock-counting and reordering but have no concept of customer relationships, campaigns, or storefronts. FashionFlow AI's inventory module is deeply integrated: a low-stock event can trigger an AI recommendation to pause a campaign promoting that item, or the AI Store Assistant can proactively recommend which slow-moving styles to discount.

2.3 Competitive Advantages

1. **Vertical specialization** — every module is modeled around fashion retail's specific needs (variants, seasons, visual merchandising).
2. **AI as a core capability, not a bolt-on** — AI features are woven into daily workflows (writing descriptions, generating designs, recommending campaigns) rather than existing as a separate "AI tab."
3. **Unified data model** — inventory, customers, campaigns, and analytics all read from and write to the same underlying data, eliminating reconciliation work.
4. **Built for multi-branch retailers** — role-based permissions and store-level scoping support single boutiques and multi-branch chains from day one.
5. **Real-time operations** — real-time order, inventory, and messaging updates (via WebSocket infrastructure) keep

every store employee working from live data.

3. User Personas

3.1 Persona: Store Owner ("Layla")

Profile: Owns a 2-branch boutique fashion chain. Not deeply technical. Wears many hats — merchandising, finance oversight, and final decision-maker on marketing spend.

- **Goals:** Grow revenue, understand which products/campaigns actually work, spend less time on admin, maintain control over pricing and discounts across branches.
- **Frustrations:** Juggling spreadsheets and multiple logins for POS, ads, and inventory. Doesn't have a designer or full-time marketer. Struggles to know what's actually profitable.
- **Daily Workflow:** Checks dashboard each morning (sales, low stock alerts), reviews AI Store Assistant recommendations, approves campaigns created by the Marketing Manager, occasionally jumps into POS during peak hours.
- **Permissions:** Full account owner — access to all modules, billing, and administration for stores they own.
- **Responsibilities:** Final approver on discounts/campaigns above a threshold, subscription/billing management, employee role assignment.

3.2 Persona: Marketing Manager ("Omar")

Profile: Runs marketing for one or more branches. Comfortable with social media and basic analytics but not a professional copywriter or designer.

- **Goals:** Launch campaigns quickly, generate ad creative without hiring an agency, prove ROI to the Store Owner.
- **Frustrations:** Writing product descriptions and ad copy takes too long. Doesn't have design skills for seasonal creative. Hard to know which segment to target.
- **Daily Workflow:** Uses AI Campaign Generator to draft ad copy and captions, schedules campaigns, monitors campaign analytics dashboard, coordinates with Store Owner on budget approval.
- **Permissions:** Full access to Marketing, Campaigns, Landing Pages, Analytics (marketing-scoped); read-only on Inventory/Finance.
- **Responsibilities:** Campaign creation and scheduling, audience segmentation, reporting campaign performance.

3.3 Persona: Sales Employee ("Mona")

Profile: Works the sales floor, handles walk-in customers and POS transactions.

- **Goals:** Process sales quickly, look up product availability across branches, help customers with returns/exchanges.
- **Frustrations:** Slow POS systems, not knowing if an item is in stock at another branch, no visibility into customer purchase history to make recommendations.
- **Daily Workflow:** Opens POS at shift start, processes walk-in orders, checks stock across branches for customer requests, processes returns/refunds within her permission threshold.
- **Permissions:** POS and Order creation, read-only Product/Inventory lookup, no access to Marketing, Analytics, or Settings.
- **Responsibilities:** Point-of-sale transactions, basic customer service, escalating refund requests above her threshold to a Manager.

3.4 Persona: Warehouse Employee ("Karim")

Profile: Manages stockroom, receives shipments, and fulfills online orders.

- **Goals:** Keep stock counts accurate, receive supplier shipments efficiently, fulfill orders on time.
- **Frustrations:** Manual stock counting, delayed visibility into incoming purchase orders, mismatches between system stock and physical stock.
- **Daily Workflow:** Receives Purchase Orders, updates inventory upon receiving stock, performs stock adjustments/counts, marks online orders as fulfilled/shipped.
- **Permissions:** Full access to Inventory, Purchase Orders, and Suppliers; Order fulfillment status updates; no access to Marketing, Customers (PII), or Analytics.
- **Responsibilities:** Inventory accuracy, receiving and reconciling purchase orders, order fulfillment.

3.5 Persona: Platform Administrator ("Anthropic-Ops" / internal)

Profile: FashionFlow AI internal staff member responsible for platform health, tenant support, and compliance.

- **Goals:** Ensure platform uptime, support merchants with account issues, monitor for abuse/fraud, manage subscription plans at the platform level.
- **Frustrations:** Needs efficient tools to investigate merchant issues without violating data privacy norms.
- **Daily Workflow:** Monitors platform health dashboards (Horizon, Pulse, Sentry), handles escalated support tickets, manages store suspensions, uses controlled impersonation to debug merchant-reported issues.
- **Permissions:** Cross-tenant administrative access, scoped and audit-logged; cannot silently access billing/payment credentials.
- **Responsibilities:** Platform-wide administration, tenant support, compliance and audit oversight, subscription plan configuration.

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4. User Roles & Permission Matrix

4.1 Role Definitions

Role	Scope	Description
Platform Admin	Cross-tenant	Anthropic-side/internal staff. Manages the platform itself, not an individual store's day-to-day operations.
Store Owner	Single account (may own multiple stores/branches)	Full control over all stores they own, billing, and staff.
Manager	One or more branches (assigned)	Elevated operational access within assigned branches; cannot manage billing or delete the store.
Employee	One branch (assigned)	Limited to operational tasks (POS, orders, inventory lookups) depending on sub-type (Sales vs. Warehouse).

Note on Employee sub-types: "Employee" is a single role at the authentication/permission-model level, but permissions are further scoped by an assigned **Department** (Sales Floor, Warehouse, Customer Support). This avoids proliferating full role types while still allowing fine-grained restriction. Store Owners configure department-based permission profiles from Settings.

4.2 Ownership Rules

- A **Store** is always owned by exactly one Store Owner account (the original registrant or a transferred owner).
- Ownership transfer requires explicit confirmation from both the outgoing and incoming owner, and is logged in the audit log.
- A Store Owner account may own multiple Stores (multi-branch or multi-brand); each Store has independent settings, catalog, and staff, but can optionally share a customer database (configurable at account level — see Business Rules §11.9).
- Deleting a Store Owner account requires either transferring or deactivating all owned Stores first (see Business Rules §11.11 on deletion/cascade rules).

4.3 Permission Matrix (Module-Level)

Legend: **F** = Full (Create/Read/Update/Delete), **W** = Write (Create/Read/Update, no delete), **R** = Read-only, **X** = No access, **C** = Configurable by Store Owner

Module	Platform Admin	Store Owner	Manager	Employee (Sales)	Employee (Warehouse)
Store Settings	R (support only)	F	R	X	X
Billing & Subscription	R (support only)	F	X	X	X
Products & Catalog	X	F	F	R	R
Categories/Brands/Collections	X	F	F	R	R
Inventory & Stock	X	F	F	R	F
Suppliers & Purchase Orders	X	F	F	X	F
Orders (Online)	X	F	F	W (create/update status)	W (fulfillment only)
POS (Walk-in)	X	F	F	F	X
Customers / CRM	X	F	F	R (own interactions)	X
Messaging Center	X	F	F	W	X
Marketing & Campaigns	X	F	C (default: F)	X	X
Landing Pages / Storefront	X	F	C (default: W)	X	X
Analytics & Reports	X	F	F (branch-scoped)	X	X
AI Features	X	F	C (default: W)	X	X
Notifications (own)	F	F	F	F	F
User & Role Management	X (platform users only)	F	W (within branch, cannot create Managers)	X	X
Platform Administration	F	X	X	X	X

4.4 Access Restriction Rules

- A **Manager** can be granted or restricted from Marketing, Landing Page, and AI feature access on a per-branch basis by the Store Owner; defaults are shown above but are configurable (**C**).
- Managers cannot create or promote another user to **Manager** or **Store Owner** — only the Store Owner can assign the Manager role.
- An **Employee (Sales)** cannot process a refund above a configurable threshold (default: platform default of local-currency equivalent of $100) without Manager or Store Owner approval (see Business Rules §11.5).
- An **Employee (Warehouse)** has no access to customer PII (names/contact info are masked in fulfillment views, showing only shipping label data required to fulfill the order).
- **Platform Admin** access to a merchant's store data is read-only by default and restricted to what is necessary for support; any deeper access (e.g., impersonation) requires explicit session logging and is time-boxed (see §4.5).

4.5 Admin Impersonation

Platform Admins may impersonate a Store Owner or staff account strictly for support/debugging purposes, subject to:

- Impersonation sessions are time-boxed (default: 30 minutes, extendable with re-justification).
- Every impersonation session is fully audit-logged (admin identity, target account, timestamp, actions taken).
- Impersonated sessions are visually indicated in the UI (a persistent banner) to prevent silent impersonation.
- Impersonation cannot be used to change billing/payment method details or to export customer PII in bulk.

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5. Complete Business Workflow

5.1 End-to-End Lifecycle

```
Store Registration
      ↓
Onboarding (business profile, branding, currency, timezone)
      ↓
Store Setup (branches, staff invites, payment methods, tax settings)
      ↓
Products (catalog, categories, brands, variants)
      ↓
Inventory (stock levels, suppliers, purchase orders)
      ↓
Customers (profiles created via orders, CRM enrichment, segmentation)
      ↓
Orders (online + POS, fulfillment, refunds)
      ↓
Marketing (audience built from CRM segments + catalog)
      ↓
Campaigns (scheduled, sent, tracked)
      ↓
Analytics (sales, campaign, inventory, customer dashboards)
      ↓
AI (content generation, forecasting, recommendations feed back into every stage above)
      ↓
Business Growth (repeat purchase, retention, expansion to new branches)
```

5.2 Stage-by-Stage Description

5.2.1 Store Registration

A prospective merchant signs up with email/password or social login (Google/Facebook). A verified email is required before proceeding. This creates a **Store Owner** account and an initial (unconfigured) **Store**.

5.2.2 Onboarding

A guided wizard collects: business name, business category confirmation (fashion vertical), branding (logo, primary color), currency, timezone, and initial subscription plan selection. Onboarding is resumable — a merchant can leave and return without losing progress.

5.2.3 Store Setup

The merchant configures branches (if multi-branch), invites staff (Managers/Employees) with role assignment, connects a payment gateway, and configures tax rules. A Store is not eligible to accept live orders until at minimum one payment method is configured (see Business Rules §11.2).

5.2.4 Products

Merchandising begins: categories, brands, and collections are created, then products with variants (size/color combinations), pricing, and images are added. Products can be created manually, via CSV import, or drafted with AI assistance (AI Product Writer).

5.2.5 Inventory

Each product variant carries its own stock count. Initial stock can be entered manually, imported via CSV, or established through a Purchase Order received from a Supplier. Low-stock thresholds are configured per-variant or globally.

5.2.6 Customers

Customer profiles are created automatically on first purchase (online or POS) or can be manually added by staff. Over time, customer profiles accumulate purchase history, enabling automatic segmentation (New, Frequent, VIP, Inactive).

5.2.7 Orders

Orders originate from two channels — the **Public Storefront** (online) or **POS** (walk-in) — and converge into a single Order Management system with a unified status pipeline, inventory deduction, and audit trail.

5.2.8 Marketing

Marketing draws on the product catalog and customer segments to build targeted audiences. Marketing Managers (or the Store Owner) create campaigns, optionally assisted by the AI Campaign Generator.

5.2.9 Campaigns

Campaigns are scheduled and sent through configured channels (WhatsApp, Email, and on-platform storefront placements/landing pages). Campaign performance (opens, clicks, conversions, revenue attributed) is tracked back to Orders.

5.2.10 Analytics

All of the above stages emit data consumed by the Analytics module: sales trends, product performance, customer lifetime value, campaign ROI, and inventory health — giving the Store Owner a single pane of glass.

5.2.11 AI

The AI module is not a separate final stage but a cross-cutting capability: it writes product content at the Products stage, forecasts stock needs at the Inventory stage, generates campaign creative at the Marketing stage, and surfaces proactive recommendations (via the AI Store Assistant) at every stage based on Analytics data.

5.2.12 Business Growth

The cumulative effect — better content, targeted campaigns, smarter inventory decisions, and retained customers — is measured through the KPIs defined in §1.6, closing the loop back into Analytics.

5.3 Cross-Module Interactions

Trigger	Source Module	Effect	Target Module
Order placed	Orders/POS	Stock decremented; customer profile updated	Inventory, CRM
Stock falls below threshold	Inventory	Low-stock notification sent; AI may suggest pausing related campaigns	Notifications, AI, Marketing
Campaign launched	Marketing	Real-time progress notification; analytics tracking begins	Notifications, Analytics
Customer inactive 60+ days	CRM	Segment reclassified to "Inactive"; may trigger win-back campaign suggestion	AI, Marketing
Purchase Order received	Suppliers	Inventory stock increased; transaction logged	Inventory
Refund processed	Orders	Stock optionally restocked; customer LTV recalculated	Inventory, CRM, Analytics

6. Product Modules

This chapter documents every module in FashionFlow AI in full detail. Each module section follows a consistent structure: Purpose, Business Goals, Entities, Relationships, Screens, CRUD Operations, Filters & Search, Business Rules, Validation, Permissions, Edge Cases, Notifications, Audit Logs, Acceptance Criteria, and Future Improvements.

6.1 Authentication Module

Purpose: Securely identify and authenticate every user (Store Owner, Manager, Employee, Platform Admin) and manage account lifecycle.

Business Goals: Minimize friction at signup while ensuring account security; support social login to reduce signup abandonment; protect merchant and customer data.

Entities: `User`, `Session`, `OtpCode`, `RecoveryCode`, `SocialAccount`, `LoginAttempt`.

Relationships: A `User` belongs to one `Store Owner account` (directly, or as staff of a Store); a `User` has many `Sessions`; a `User` optionally has many `SocialAccount` links (Google, Facebook).

Screens: Sign Up, Log In, Verify Email, Verify Phone (OTP), Forgot Password, Reset Password, Two-Factor Setup, Two-Factor Challenge, Recovery Codes, Active Sessions, Account Deletion Confirmation.

CRUD Operations:

Operation	Entity	Notes
Create	User	Via signup form or social OAuth
Read	Session	List of active sessions with device/location
Update	User	Password change, phone number, 2FA toggle
Delete	Session	Revoke a specific session or "log out everywhere"
Delete	User	Full account deletion (soft delete + scheduled hard delete, see §11.11)

Business Rules:

- Email must be verified before a Store Owner can accept live payments or publish a storefront.
- Phone verification (OTP) is required before enabling SMS/WhatsApp-based notifications for a store.
- Two-Factor Authentication (2FA) is optional for Employees/Managers but **mandatory** for Store Owners once the store has processed its first live payment (protects financial access).
- Recovery codes are generated once at 2FA setup; regenerating invalidates all previous codes.
- A maximum of 5 failed login attempts within 15 minutes triggers a temporary lockout (15 minutes) and an email alert to the account.

Validation Rules:

- Password: minimum 8 characters, at least one number and one letter.
- Email: standard RFC email format, uniqueness enforced per platform (not per store).
- Phone: validated against E.164 format prior to OTP dispatch.
- OTP codes: 6 digits, expire after 10 minutes, single-use.

Permissions: All roles can manage their own authentication settings. Only Store Owners can view/manage 2FA enforcement policy for their staff. Platform Admins can force-reset a compromised account's credentials (audit-logged).

Edge Cases:

- User attempts social login with an email already registered via password: system prompts to link accounts after password verification, rather than creating a duplicate account.
- OTP requested repeatedly within a short window: rate-limited to 1 request per 60 seconds per phone number.
- User deletes account while owning active Stores: deletion is blocked until Stores are transferred or deactivated (see §11.11).
- Session revoked mid-request: in-flight request completes, subsequent requests are rejected with re-authentication required.

Notifications: Email — new device login, password changed, 2FA enabled/disabled, account deletion requested.

WhatsApp/SMS — OTP codes only.

Audit Logs: All authentication-sensitive events (login, password change, 2FA change, impersonation) are logged with timestamp, IP, device fingerprint, and outcome.

Acceptance Criteria: - [ ] User can sign up via email/password or Google/Facebook OAuth. - [ ] Email verification is enforced before live payment activation. - [ ] 2FA can be enabled and enforced per the rules above. - [ ] Account deletion is blocked while Stores are owned and unresolved. - [ ] All failed and successful authentication events are audit-logged.

Future Improvements: Passkey/WebAuthn support; SSO for Enterprise tier; biometric login on mobile app.

6.2 Store Management Module

Purpose: Manage the core identity, branding, and configuration of each Store (and its branches).

Business Goals: Give merchants full control of brand presentation and operational settings without engineering involvement.

Entities: `Store`, `Branch`, `Branding`, `SubscriptionPlan`, `StoreSettings`, `SocialAccountLink`.

Relationships: A `Store` has many `Branch` records (for multi-branch retailers); a `Store` has one `Branding` profile and one active `SubscriptionPlan`; a `Store` has many `StoreSettings` (currency, timezone, notification preferences).

Screens: Store Profile, Branding Settings, Branch Management, Subscription & Billing, Notification Preferences, Connected Social Accounts.

CRUD Operations: Full CRUD on `Branch` (Store Owner/Manager per permission matrix); Update on `Store` profile/branding; Read/Update on `SubscriptionPlan` (upgrade/downgrade); Create/Delete on `SocialAccountLink`.

Filters & Search: Branch list filterable by status (active/inactive) and region.

Business Rules:

- A Store must have at least one active Branch at all times; the last remaining Branch cannot be deleted (must be deactivated via full store closure instead).
- Currency and timezone are set at Store level and inherited by all Branches; changing currency after transactions exist requires explicit confirmation and does not retroactively convert historical figures (historical reports display the currency active at the time of transaction).
- Subscription downgrades that would exceed the new plan's limits (e.g., branch count, staff count) are blocked until the merchant brings usage within the new plan's limits.

Validation Rules: Store name: 2–100 characters. Branch address: required fields (country, city, address line) validated against a supported country list.

Permissions: Store Owner — full control. Manager — read-only on Store profile; can update Branch-level operational settings (hours, contact) if granted.

Edge Cases: Merchant attempts to delete a Branch with open (unfulfilled) Orders — blocked with an explicit error listing the blocking orders. Currency change mid-billing-cycle — subscription is billed in the currency active at the start of that billing cycle.

Notifications: Subscription renewal reminders, payment failure alerts, branch deactivation confirmations.

Audit Logs: All Store/Branch configuration changes logged with before/after values.

Acceptance Criteria: - [ ] Store Owner can create, edit, and deactivate Branches (except the last one). - [ ] Subscription upgrade/downgrade respects plan limits. - [ ] Branding changes propagate to storefront and landing pages without redeploy.

Future Improvements: White-label domains per branch; franchise-mode multi-owner branches.

6.3 Products Module (including Categories, Brands, Collections, Variants)

Purpose: Central catalog management for all sellable items, structured for fashion retail's variant complexity.

Business Goals: Enable fast, accurate catalog creation; support rich merchandising (categories, brands, collections) to drive discoverability and campaign targeting.

Entities: `Product`, `Category`, `Brand`, `Collection`, `Tag`, `ProductVariant`, `ProductImage`.

Relationships: - A `Product` belongs to one `Category`, optionally one `Brand`, and many `Collections` and `Tags`

(many-to-many). - A `Product` has many `ProductVariant` records (e.g., Black/M, White/L). - A `ProductVariant` has its own SKU, barcode, price override (optional), cost, and stock count (stock lives in the Inventory module but is 1:1 per variant). - A `Product` has many `ProductImage` records; images can be assigned at the product level or per-variant (e.g., a color-specific photo).

Example:

Product	Variant	SKU	Stock
Oversized T-Shirt	Black / M	TS-BLK-M	24
Oversized T-Shirt	Black / L	TS-BLK-L	18
Oversized T-Shirt	White / M	TS-WHT-M	30
Oversized T-Shirt	White / XL	TS-WHT-XL	5 (low stock)

Screens: Product List, Product Create/Edit, Variant Matrix Editor, Category Manager, Brand Manager, Collection Manager, Bulk Import (CSV), Bulk Export.

CRUD Operations: Full CRUD on Product, Category, Brand, Collection, Variant. Images support multi-upload, reorder, and delete.

Filters & Search: Filter by Category, Brand, Collection, Tag, Status (Draft/Active/Archived), Stock status (In Stock/Low/Out of Stock), Price range. Full-text search across product title, SKU, and tags.

Business Rules:

- A Product cannot be published (status = Active) without at least one Variant carrying a price and a valid SKU.
- SKU must be unique within a Store (not globally across the platform).
- Deleting a Category that has assigned Products is blocked; the merchant must reassign or bulk-move Products first, or the system offers to move them to an "Uncategorized" default category.
- Archiving a Product hides it from the Public Storefront but preserves historical Order line-item references (soft delete pattern — see §11.10).
- A Product's "Draft" status is never visible on the storefront, regardless of stock levels.
- Discount pricing (if set) must always be less than the base price; system rejects a discount price ≥ base price.

Validation Rules:

- Product title: 3–150 characters, required.
- Price: must be a positive number with up to 2 decimal places (or 0 decimal places for zero-decimal currencies).
- SKU: alphanumeric, 3–40 characters, unique per store.
- Barcode: optional, validated against standard formats (UPC/EAN) if provided, not enforced as unique across stores.
- At least one image is required before a Product can move from Draft to Active.

Permissions: Store Owner/Manager — full CRUD. Sales Employee/Warehouse Employee — read-only, no price/description edit access.

Edge Cases:

- Bulk CSV import contains a duplicate SKU already in the system: the row is flagged as an error and skipped; the rest of the import proceeds (partial import with an error report downloadable at the end).
- A Variant's stock reaches 0 while it's featured in an active Campaign: system does not auto-remove it from the campaign, but flags it in the AI Store Assistant's recommendations and on the campaign dashboard as "sold out — review".
- Merging near-duplicate products (e.g., accidental double-entry): supported via a manual "merge products" action that reconciles variants and reassigns historical order references.

Notifications: Product published successfully, bulk import completed (with error summary), low-stock threshold reached (see Inventory module).

Audit Logs: All price changes, status changes (Draft/Active/Archived), and deletions are logged with the acting user.

Acceptance Criteria: - [ ] A Product can be created with 1–N variants across size/color combinations without manual SKU duplication. - [ ] A Product cannot be set Active without a valid price, SKU, and at least one image. - [ ] CSV bulk import supports partial success with a downloadable error report. - [ ] Deleting a Category with assigned Products is blocked or offers reassignment.

Future Improvements: AI-suggested categorization on import; automatic duplicate-product detection; 3D/AR product

previews.

6.4 Inventory Module

Purpose: Track and manage stock levels per variant across all branches, and manage the supply chain that replenishes it.

Business Goals: Prevent overselling, minimize stockouts on popular items, and give merchants visibility into inventory health and value.

Entities: `StockLevel` (per Variant, per Branch), `InventoryAdjustment`, `InventoryTransaction`, `LowStockAlert`.

Relationships: A `ProductVariant` has one `StockLevel` per `Branch` (multi-branch stock is tracked independently, with an optional "shared pool" mode — see Business Rules §11.9). Every stock change (sale, return, adjustment, purchase order receipt) creates an `InventoryTransaction` for full traceability.

Screens: Stock Overview (per branch and aggregate), Adjustment Form, Transaction History, Low Stock Alerts, Inventory Valuation Report.

CRUD Operations: Create/Read on `InventoryTransaction` (immutable — never edited or deleted, only appended, to preserve audit integrity); Update on `StockLevel` via Adjustments (never direct edits — all changes go through the Adjustment flow so a reason and audit trail are always captured).

Filters & Search: Filter by Branch, Product, Category, Stock status (In Stock/Low/Out of Stock/Overstock), Transaction type (Sale/Return/Adjustment/Purchase Order Receipt).

Business Rules:

- Stock cannot go negative under normal sales flow; an order attempting to sell more than available stock is blocked at checkout/POS (see Orders module edge cases for backorder exception).
- Every stock change must be attributable to a `InventoryTransaction` type: `sale`, `return`, `manual_adjustment`, `purchase_order_receipt`, `transfer`.
- Low-stock threshold defaults to 5 units per variant but is configurable globally or per-product.
- "Out of Stock" status is automatically applied when stock = 0 and automatically cleared when stock > 0 again (no manual override needed, though manual override is possible for pre-order scenarios — see §11.3).
- Inter-branch stock transfers create two linked `InventoryTransaction` records (a `transfer_out` at the source branch and `transfer_in` at the destination), keeping both branches' histories independently auditable.

Validation Rules: Adjustment quantity: non-zero integer (positive or negative). Adjustment reason: required free-text or a reason-code selection (Damaged, Lost, Recount, Other).

Permissions: Store Owner/Manager — full access including manual adjustments. Warehouse Employee — full access to adjustments and purchase order receiving. Sales Employee — read-only stock lookup only.

Edge Cases:

- Two POS terminals attempt to sell the last unit of a variant simultaneously: the platform uses row-level locking/optimistic concurrency on `StockLevel`; the second transaction is rejected in real time with a "no longer available" message rather than allowing negative stock (see §11.12 Concurrency).
- A Purchase Order is partially received (e.g., 80 of 100 units arrive): the system supports partial receiving, keeping the Purchase Order in a "Partially Received" status until the remainder arrives or the order is manually closed.
- A returned item is damaged and should not be restocked: the Return flow allows marking a returned unit as "damaged — do not restock," which logs an `InventoryTransaction` of type `return` with zero stock impact plus a `manual_adjustment` note.

Notifications: Low stock alert (in-app, email), out-of-stock alert, purchase order received confirmation.

Audit Logs: Every `InventoryTransaction` is itself the audit record — immutable, timestamped, and attributed to the acting user or system process (e.g., "Order #1234").

Acceptance Criteria: - [ ] Stock cannot go negative through the standard sales flow. - [ ] Every stock change is traceable to a specific transaction type and actor. - [ ] Low-stock alerts fire at the configured threshold and are visible in Notifications and the Analytics dashboard. - [ ] Concurrent sale attempts on the last unit are resolved without overselling.

Future Improvements: AI-driven automatic reorder point suggestions (see AI Inventory Forecasting, §7.6); barcode-scanner-based stock counts; RFID integration for large warehouses.

6.5 Suppliers & Purchase Orders Module

Purpose: Manage the merchant's supply chain — who they buy from and what/when they've ordered.

Business Goals: Streamline restocking, maintain supplier relationships and pricing history, and ensure incoming stock is reconciled accurately.

Entities: `Supplier`, `PurchaseOrder`, `PurchaseOrderLineItem`.

Relationships: A `Supplier` has many `PurchaseOrder` records. A `PurchaseOrder` has many `PurchaseOrderLineItem` records, each referencing a `ProductVariant` and an ordered quantity/cost.

Screens: Supplier List, Supplier Profile, Purchase Order List, Purchase Order Create/Edit, Receiving Screen.

CRUD Operations: Full CRUD on Supplier and Purchase Order while in "Draft" status. Once a Purchase Order is marked "Sent" to the supplier, line items become locked (quantity/cost cannot be edited — only cancelled and re-created) to preserve an accurate record of what was actually ordered.

Filters & Search: Filter Purchase Orders by Supplier, Status (Draft/Sent/Partially Received/Received/Cancelled), Branch, Date range.

Business Rules:

- A Purchase Order's status automatically transitions to "Partially Received" or "Received" based on receiving actions in the Inventory module (see §6.4).
- Cancelling a Purchase Order that has been partially received is allowed, but the already-received stock remains in inventory (cancellation only voids the outstanding, unreceived portion).
- Supplier records cannot be deleted if they have associated Purchase Orders; they can be archived instead.

Validation Rules: Purchase Order must have at least one line item with a positive quantity before it can be moved out of Draft status. Supplier contact email (if provided) must be valid format.

Permissions: Store Owner/Manager — full access. Warehouse Employee — full access to Purchase Orders and receiving; cannot see supplier cost/pricing terms if the Store Owner restricts financial visibility (configurable).

Edge Cases: Supplier sends more units than ordered — receiving screen allows recording the actual received quantity, which may exceed the ordered quantity, with a flag for merchant review. A Purchase Order line item's product is deleted from the catalog after the PO is sent — the line item retains a snapshot of the product name/SKU at time of ordering so historical POs remain legible.

Notifications: Purchase Order sent confirmation, goods received confirmation, overdue Purchase Order reminder (configurable expected-delivery date passed without receipt).

Audit Logs: All Purchase Order status transitions and receiving actions logged with acting user and timestamp.

Acceptance Criteria: - [ ] Purchase Orders progress through Draft → Sent → Partially Received/Received → Closed correctly. - [ ] Receiving stock through a Purchase Order updates Inventory `StockLevel` and creates the correct `InventoryTransaction`. - [ ] Suppliers with existing Purchase Orders cannot be hard-deleted.

Future Improvements: Supplier portal for direct order confirmation; automated reordering based on AI forecasting; landed-cost calculation (freight/duties).

6.6 Orders & POS Module

Purpose: Manage the full lifecycle of a sale, whether initiated online or in-person, from creation through fulfillment, payment, and potential refund.

Business Goals: Provide a single, unified order pipeline regardless of sales channel; minimize checkout friction; ensure accurate financial and inventory reconciliation.

Entities: `Order`, `OrderLineItem`, `OrderStatusHistory`, `Payment`, `Refund`, `Coupon`, `Invoice`.

Relationships: An `Order` has many `OrderLineItem` records (each referencing a `ProductVariant`, quantity, and price at time of sale). An `Order` has one or more `Payment` records (supporting split payments) and zero or more `Refund` records. An `Order` optionally references a `Coupon` applied at checkout, and generates one `Invoice`.

Screens: Order List, Order Detail/Timeline, POS Terminal (product grid, cart, payment), Refund Flow, Coupon Manager, Invoice View/Download.

CRUD Operations: Create Order (online checkout or POS); Read/Update Order status; Create Refund (against an existing Order); Full CRUD on Coupons.

Filters & Search: Filter Orders by Status (Pending/Processing/Fulfilled/Cancelled/Refunded), Channel (Online/POS), Branch, Date range, Customer. Search by Order number or customer name.

Order Status Pipeline:

Status	Meaning	Who can set it
Pending	Order placed, payment not yet confirmed	System (on checkout)
Processing	Payment confirmed, awaiting fulfillment	System (on payment webhook) / Staff
Fulfilled	Item handed over (POS) or shipped (online)	Staff (Warehouse for online, Sales for POS)
Cancelled	Order voided before fulfillment	Staff (with permission)
Refunded	Fully or partially refunded post-fulfillment	Staff (with refund permission)

Business Rules:

- Stock is decremented at the point an Order moves to "Processing" (payment confirmed) for online orders, and immediately at sale completion for POS orders — never at "Pending," to avoid holding stock hostage to abandoned/unpaid carts.
- A Coupon cannot be combined with another Coupon in the same Order (no stacking) unless the Coupon is explicitly flagged as "stackable" by the merchant (see Business Rules §11.4 for full discount-stacking logic).
- Refunds cannot exceed the original payment amount of the Order (partial refunds are supported and tracked cumulatively against the original total).
- A refund automatically triggers an `InventoryTransaction` of type `return`, restoring stock — unless the staff member explicitly marks the returned item as damaged/non-restockable.
- Orders placed via POS are attributed to the Branch and staff member who processed them; online orders are attributed to the Branch nearest the customer (if multi-branch fulfillment is enabled) or a default fulfillment branch.
- Taxes are calculated based on the Branch's configured tax rate at the time of sale (not retroactively recalculated if tax rates change later).

Validation Rules: An Order must have at least one Line Item. Refund amount must be > 0 and ≤ remaining refundable balance. Coupon codes: case-insensitive matching, must be active and within its validity date range.

Permissions: Store Owner/Manager — full access to all Orders and unrestricted refunds. Sales Employee — can create POS Orders and process refunds up to their configured threshold (default $100 equivalent); above-threshold refunds are routed to Manager approval. Warehouse Employee — can update fulfillment status only, no refund access, no pricing visibility beyond what's needed to pack the order.

Edge Cases:

- Customer requests a refund for an item no longer in the catalog (deleted product): refund is still processable because `OrderLineItem` stores a snapshot of the product name/price/SKU at time of sale, independent of the live catalog record.
- Partial refund followed by a second partial refund: system tracks cumulative refunded amount and blocks any refund attempt that would exceed the original payment total.
- Payment gateway webhook for a Pending order arrives after the customer has already abandoned/left the checkout page: Order still transitions to "Processing" correctly since state transitions are driven by webhook events, not client-side state.
- Attempted oversell at POS (stock changed between page load and checkout, e.g., another branch's online order took the last unit in a shared pool): POS blocks the sale with a real-time "insufficient stock" error.

Notifications: Order confirmation (customer + staff), payment failed, order fulfilled/shipped (customer), refund processed (customer), new order alert (staff, real-time via WebSocket).

Audit Logs: Every status transition, refund, and coupon application is logged with acting user/system and timestamp; the Order Detail screen renders this as a human-readable timeline.

Acceptance Criteria: - [ ] Orders correctly progress through the status pipeline for both Online and POS channels. - [ ] Stock is decremented at the correct pipeline stage per channel. - [ ] Refunds respect the cumulative-total validation rule and staff refund thresholds. - [ ] Every Order has a complete, human-readable timeline of all status changes.

Future Improvements: Buy-online-pickup-in-branch (BOPIS) flow; layaway/installment payment plans; automated fraud-risk scoring on high-value orders.

6.7 Customers / CRM Module

Purpose: Maintain a complete profile of every customer, their purchase history, and their engagement segment.

Business Goals: Enable personalized marketing and service by giving every staff member visibility into who a customer is and what they've bought.

Entities: `Customer`, `CustomerAddress`, `CustomerNote`, `CustomerSegmentAssignment`.

Relationships: A `Customer` has many `Order` records (purchase history), many `CustomerAddress` records (address book), and many `CustomerNote` records (staff-authored). A `Customer` has one active `CustomerSegmentAssignment` at a time, recalculated periodically (see Business Rules §11.6).

Screens: Customer List, Customer Profile (with purchase history, notes, LTV), Segment Manager, Address Book.

CRUD Operations: Full CRUD on Customer profile and Address Book; Create/Read on Notes (notes are not editable after creation beyond a short grace window, to preserve integrity of staff communication history — they can be deleted by a Manager/Owner but not silently altered).

Filters & Search: Filter by Segment (New/Frequent/VIP/Inactive), Branch (first purchase location), Lifetime Value range, Last purchase date range. Search by name, email, or phone.

Segments:

Segment	Default Definition
New	First purchase within the last 30 days, fewer than 2 total orders
Frequent	3+ orders within the last 12 months
VIP	Lifetime value in the top 10% of the store's customer base, or manually tagged
Inactive	No purchase in the last 90 days, having previously purchased at least once

Business Rules:

- Segment assignment is recalculated nightly via a scheduled job; it is also viewable as of the last calculation time so staff know the data's freshness.
- A customer manually tagged "VIP" by staff retains that tag regardless of the automatic LTV-percentile calculation, until manually removed.
- Customer Lifetime Value (LTV) is calculated as the sum of all non-refunded Order totals attributed to the customer.
- Deleting a Customer profile (e.g., for privacy/GDPR-style requests) anonymizes rather than deletes associated Orders (preserving financial/tax records) — see Business Rules §11.11.

Validation Rules: Email or phone is required (at least one) to create a Customer profile. Address fields validated against supported country list.

Permissions: Store Owner/Manager — full access. Sales Employee — read-only profile access plus ability to add Notes; no access to full purchase-value analytics beyond what's shown on the profile. Warehouse Employee — no access (PII-restricted, per §4.4).

Edge Cases:

- A walk-in customer makes a purchase without providing contact info: an anonymous/guest Order is recorded without creating a Customer profile; if the same person later provides contact info on a subsequent visit, no automatic merge occurs (manual merge tool available to Managers).
- Two Customer profiles are discovered to represent the same person (duplicate entries): a manual "merge customers" tool combines Order history and recalculates LTV/segment.

Notifications: New customer created (optional, staff-facing), segment change to VIP (optional, celebratory internal notification), customer requests data deletion (compliance alert to Store Owner).

Audit Logs: Profile edits, note additions/deletions, and merges are logged.

Acceptance Criteria: - [ ] Customer profiles automatically aggregate purchase history and LTV. - [ ] Segments are recalculated on schedule and are manually overridable for VIP tagging. - [ ] Deletion requests anonymize rather than destroy financial records.

Future Improvements: Loyalty points program; predictive segment transitions (AI-driven, see §7.7); customer self-service portal.

6.8 Marketing & Campaigns Module

Purpose: Enable merchants to plan, create, target, schedule, and measure promotional campaigns.

Business Goals: Drive revenue through targeted promotions while giving merchants ROI visibility they'd otherwise need a dedicated marketer to produce.

Entities: `Campaign`, `CampaignAudience`, `PromoCode`, `CampaignSchedule`, `CampaignBudget`.

Relationships: A `Campaign` targets one `CampaignAudience` (built from Customer Segments and/or custom filters), references one or more Products/Collections being promoted, optionally includes a `PromoCode`, and has one `CampaignSchedule` and one `CampaignBudget`.

Campaign Types: Product Campaign, Seasonal Campaign, Flash Sale, Discount Campaign.

Screens: Campaign List, Campaign Builder (audience → content → schedule → budget), Promo Code Manager, Campaign Performance Dashboard.

CRUD Operations: Full CRUD on Campaigns while in "Draft" status. Once "Scheduled" or "Active," core targeting/content is locked (to preserve reporting integrity) but the campaign can be paused or cancelled.

Filters & Search: Filter by Type, Status (Draft/Scheduled/Active/Completed/Cancelled/Paused), Channel, Date range.

Business Rules:

- A Flash Sale campaign requires a defined end time (cannot run indefinitely) to preserve the urgency mechanic; the system enforces a maximum duration (configurable, default 7 days).
- Discount stacking between a Campaign's promo code and an existing product-level discount price follows the priority rules defined in Business Rules §11.4 (in short: the system applies whichever single discount benefits the customer most, unless "stackable" is explicitly enabled).
- A Campaign cannot target a Customer Segment that would result in zero eligible recipients; the system warns before scheduling send.
- Budget is a soft cap for ad-spend tracking (when integrated with Meta/Google ad accounts) — the platform does not enforce hard spend cutoffs itself but surfaces spend-vs-budget in the dashboard.

Validation Rules: Campaign name required, 3–100 characters. Schedule start time must be in the future at creation (cannot schedule into the past). Promo code: alphanumeric, 3–20 characters, unique per store, case-insensitive.

Permissions: Store Owner — full access. Manager — full access by default, configurable to restrict campaign launch to Store Owner approval above a budget threshold. Sales/Warehouse Employees — no access.

Edge Cases:

- A campaign is scheduled to promote a Product that goes out of stock before send time: the campaign still sends (to avoid silently failing a scheduled marketing push) but the AI Store Assistant flags it and the merchant is notified to intervene manually.
- Two campaigns simultaneously target overlapping audiences with conflicting discounts: resolved via Campaign Priority rules (§11.7) — higher-priority campaign's discount applies; the customer is never shown or charged both.

Notifications: Campaign scheduled confirmation, campaign sent confirmation, campaign performance summary (24 hours post-send), low audience size warning.

Audit Logs: Campaign creation, edits (while in Draft), and status transitions are logged.

Acceptance Criteria: - [ ] Campaigns can be built end-to-end (audience → content → schedule → budget) and launched on schedule. - [ ] Discount conflicts between simultaneous campaigns resolve deterministically per priority rules. - [ ] Campaign performance (opens, clicks, conversions, revenue) is tracked and visible within 24 hours of send.

Future Improvements: A/B testing of campaign content; automated send-time optimization (AI-driven); multi-step drip campaigns.

6.9 Landing Pages & Public Storefront Module

Purpose: Provide merchants a customer-facing storefront and campaign-specific landing pages without needing a web developer.

Business Goals: Convert marketing traffic into sales; give brands a professional, on-brand online presence.

Entities: `Storefront`, `LandingPage`, `LandingPageTemplate`, `Cart`, `CheckoutSession`, `SEOConfig`, `CustomDomain`.

Relationships: A `Store` has one `Storefront` (the always-on catalog site) and many `LandingPage` records (campaign-specific, time-bound pages). A `LandingPage` is built from a `LandingPageTemplate` and can have its own `SEOConfig` and optional `CustomDomain`.

Screens: Storefront theme editor, Landing Page Builder (drag-and-drop sections), Template Gallery, Domain Settings, SEO Settings, Cart/Checkout (customer-facing).

CRUD Operations: Full CRUD on Landing Pages and Storefront theme configuration. Templates are read-only (platform-provided) but can be duplicated and customized.

Filters & Search: Landing Pages filterable by Status (Draft/Published/Expired), associated Campaign.

Business Rules:

- A Landing Page tied to a time-bound Campaign (e.g., a Flash Sale) automatically transitions to "Expired" and is unpublished when the campaign's end time passes, preventing stale/misleading pages from remaining live.
- Custom domains require DNS verification (a TXT record check) before going live; until verified, the page remains accessible only via the default FashionFlow AI subdomain.
- The Public Storefront cannot go live until the Store has: at least one Active product, one configured payment method, and a completed Terms & Privacy page (minimum legal completeness gate).

Validation Rules: Custom domain must be a valid, unique (platform-wide) domain not already claimed by another store. SEO meta description: max 160 characters (best-practice enforced with a warning, not a hard block).

Permissions: Store Owner — full access. Manager — configurable (default: can edit content, cannot change domain/SEO/legal pages). Employees — no access.

Edge Cases: Customer adds items to Cart, then a Variant sells out via POS before Checkout completes: Checkout re-validates stock at payment time and blocks with a clear "no longer available, please update your cart" message rather than accepting an unfulfillable order.

Notifications: Domain verification success/failure, landing page published, landing page auto-expired.

Audit Logs: Publish/unpublish actions and domain changes are logged.

Acceptance Criteria: - [ ] Storefront cannot go live until the minimum completeness gate (product, payment, legal pages) is satisfied. - [ ] Time-bound Landing Pages auto-expire correctly. - [ ] Checkout re-validates stock immediately prior to payment capture.

Future Improvements: Full visual theme customization (custom CSS); multi-language storefronts; headless storefront API for brands wanting a fully custom frontend.

6.10 Messaging Center Module

Purpose: Unify all customer communication (order-related and marketing) into a single inbox for staff.

Business Goals: Prevent missed customer inquiries across channels; give staff shared context via internal notes.

Entities: `Conversation`, `Message`, `InternalNote`, `QuickReplyTemplate`.

Relationships: A `Conversation` belongs to one `Customer` and aggregates `Message` records from multiple `Channel` types (WhatsApp, Email). A `Conversation` has many `InternalNote` records visible only to staff.

Screens: Unified Inbox, Conversation Thread, Quick Reply Manager.

CRUD Operations: Create/Read Messages (sent/received); Create/Read/Delete Internal Notes; Full CRUD on Quick Reply Templates.

Filters & Search: Filter by Channel, Status (Unread/Read/Resolved), Assigned staff member. Search by customer name or message content.

Business Rules:

- Order-related system messages (e.g., shipping confirmations) appear in the same Conversation thread as customer-initiated messages for full context, but are visually distinguished as system-generated.
- A Conversation is marked "Resolved" manually by staff; new incoming messages automatically reopen a Resolved conversation.
- Internal Notes are never visible to the customer under any channel.

Validation Rules: Message content: required, non-empty; channel-specific length limits are enforced per the connected channel's own limits (e.g., WhatsApp template constraints).

Permissions: Store Owner/Manager/Marketing-permitted roles — full access. Sales Employee — access limited to conversations related to their own processed Orders (configurable to full access). Warehouse Employee — no access.

Edge Cases: Customer replies on WhatsApp after a marketing message they weren't previously in conversation with: system creates a new Conversation thread tied to their Customer profile automatically. Message delivery fails (e.g., invalid WhatsApp number): failure status is shown inline in the thread with a retry option.

Notifications: New message received (real-time, in-app + optional push).

Audit Logs: Internal note deletions are logged (content + deleting user) since they represent removed staff context.

Acceptance Criteria: - [ ] All channels converge into a single per-customer Conversation view. - [ ] Internal Notes never leak to any customer-facing channel. - [ ] Resolved conversations reopen automatically on new incoming messages.

Future Improvements: AI-suggested reply drafts within the inbox; SMS channel support; conversation assignment/routing rules.

6.11 Notifications Module

Purpose: Deliver timely, relevant alerts to the right user through the right channel.

Business Goals: Keep staff informed in real time without overwhelming them; give merchants control over what they're notified about.

Entities: `NotificationEvent`, `NotificationPreference`, `NotificationDelivery`.

Relationships: A `NotificationEvent` (e.g., "low stock," "new order") is delivered to one or more `User` records based on their role and `NotificationPreference`, producing a `NotificationDelivery` record per channel attempted.

Screens: Notification Center (in-app feed), Notification Preferences.

CRUD Operations: Read/Update (mark read) on in-app notifications; Update on Preferences (channel opt-in/out per event type).

Filters & Search: Filter by type (Order, Inventory, Marketing, System), Read/Unread status.

Business Rules: See full breakdown in Chapter 9. Every notification type has a defined default channel set and priority; users may narrow (opt out of non-critical channels) but cannot fully disable account-security notifications (e.g., new device login).

Validation Rules: N/A (system-generated content); user preference toggles are boolean per event-type/channel pair.

Permissions: Every user manages only their own preferences. Store Owner can set org-wide default preferences for newly invited staff (staff can still override for themselves except security-critical events).

Edge Cases: A user disables Email for "New Order" but the store has no other staff with Email enabled for that event and no in-app session active: the event is still recorded in the in-app feed and surfaces on next login (no notification is ever silently dropped).

Notifications: (this module *is* the notification system — see Chapter 9 for the full catalog).

Audit Logs: Preference changes are logged.

Acceptance Criteria: - [ ] Every system event maps to a defined notification with correct default channels. - [ ] Users can manage their own channel preferences except for security-critical events. - [ ] Unread notifications persist and surface on next login even if real-time delivery was missed.

Future Improvements: Digest mode (daily/weekly summary email instead of per-event); Slack/Teams integration for staff.

6.12 Payments & Wallet Module

Purpose: Process customer payments securely and give merchants visibility into their available balance and payout schedule.

Business Goals: Ensure reliable payment capture across Online and POS channels; provide transparent payout timing for merchant trust.

Entities: `PaymentMethod` (merchant-configured gateway), `Transaction`, `Payout`, `WalletBalance`.

Relationships: A `Store` has one or more configured `PaymentMethod` records (gateway credentials). Every `Payment` (from Orders) generates a `Transaction` record affecting the Store's `WalletBalance`. Funds are periodically disbursed via

`Payout` records.

Screens: Payment Method Setup, Wallet Balance Dashboard, Transaction History, Payout History.

CRUD Operations: Create/Update/Delete on Payment Methods (Store Owner only); Read-only on Transactions, Wallet Balance, and Payouts (system-generated).

Filters & Search: Filter Transactions by type (Sale/Refund/Fee/Payout), date range, Branch.

Business Rules:

- Wallet balance reflects net of platform fees, refunds, and pending holds; merchants see a clear breakdown (gross sales − fees − refunds = available balance).
- Payouts follow a configurable schedule (default: weekly) and require a verified payout destination (bank account) before the first payout can be released.
- A refund issued against a Transaction that has already been paid out is deducted from the *next* payout rather than reversing the historical payout record.

Validation Rules: Payment gateway credentials are validated via a test transaction before activation. Payout destination account details validated per supported country/bank format.

Permissions: Store Owner — full access. Manager/Employees — no access to payment configuration or wallet balance (financial data restricted to ownership level unless explicitly delegated).

Edge Cases: Payment gateway webhook confirms a payment after the customer's Order was auto-cancelled due to a timeout: the system reconciles by re-opening the Order rather than leaving an orphaned payment. Payout destination fails validation (e.g., bank rejects account): payout is held and merchant is notified to correct details, subsequent payout cycles accumulate until resolved.

Notifications: Payout sent, payout failed, payment method expiring soon, payment gateway disconnected.

Audit Logs: All payment method changes and payout events are logged.

Acceptance Criteria: - [ ] Wallet balance accurately reflects sales, fees, and refunds in real time. - [ ] Payouts are blocked until a verified destination is on file. - [ ] Refunds against already-paid-out transactions are deducted from future payouts, never reversing history.

Future Improvements: Instant payout option (for a fee); multi-currency wallet support; embedded merchant lending based on wallet history.

6.13 Reports & Analytics Module

(Full dashboard-by-dashboard detail is provided in Chapter 8; this section covers the module's structural requirements.)

Purpose: Aggregate data across every module into actionable dashboards and exportable reports.

Business Goals: Give merchants decision-quality insight without needing a data analyst.

Entities: `ReportDefinition`, `ReportExport`, `DashboardWidget`, `MetricSnapshot` (pre-aggregated for performance).

Relationships: A `DashboardWidget` reads from one or more `MetricSnapshot` aggregates (refreshed on a schedule, not computed live on every page load, for performance at scale).

Screens: Dashboard Home, Sales Report, Customer Report, Campaign Report, Inventory Report, Financial Report, Custom Report Builder.

CRUD Operations: Read-only on all standard dashboards; Create/Read/Delete on Custom Report Builder configurations and scheduled Exports.

Filters & Search: Universal date-range filter, Branch filter, comparison-period toggle (e.g., "vs. previous period").

Business Rules: Metric snapshots refresh on a defined cadence (real-time counters for today's sales; hourly aggregation for historical trend charts, to balance freshness with system load). Exported reports (CSV/PDF) reflect the data as of export time and are timestamped accordingly.

Validation Rules: Custom date ranges cannot exceed 24 months in a single query (performance guardrail); longer historical analysis requires multiple exports.

Permissions: Store Owner — full access to all reports. Manager — branch-scoped access. Employees — no access (per §4.3).

Edge Cases: A report is requested for a date range spanning a currency change: figures are shown in original transaction currency per period, with a clear note rather than a misleading blended conversion (see Store Management

§11 rules on currency).

Notifications: Scheduled report export ready for download.

Audit Logs: Report exports (who exported what, when) are logged for data-governance purposes.

Acceptance Criteria: - [ ] All dashboards load using pre-aggregated snapshots within acceptable performance thresholds (see §13). - [ ] Exports are timestamped and reflect a consistent point-in-time snapshot. - [ ] Manager-level access is correctly scoped to their assigned branch(es) only.

Future Improvements: Natural-language report queries via AI Store Assistant; scheduled recurring email reports; benchmark comparisons against anonymized industry data.

6.14 Settings & Administration Module

Purpose: Central configuration hub for store-level preferences and platform-level administrative controls.

Business Goals: Give merchants self-service control over operational settings; give Platform Admins the tools to run the business safely and support merchants effectively.

Entities (Store-level Settings): `TaxRule`, `ShippingRule`, `RefundPolicy`, `StaffPermissionProfile`. **Entities (Platform Administration):** `PlatformSubscriptionPlan`, `StoreSuspension`, `SupportTicket`, `PlatformAuditLog`.

Screens (Store-level): Tax Settings, Shipping Settings, Refund Policy Editor, Staff & Roles. **Screens (Platform-level):** Store Directory, Subscription Plan Manager, Suspension Tool, Support Ticket Queue, Platform Audit Log Viewer.

CRUD Operations: Store Owner — full CRUD on all Store-level settings entities. Platform Admin — full CRUD on Platform-level entities; read-only access into merchant Store data for support purposes (per §4.5 impersonation rules).

Filters & Search: Store Directory (Platform Admin view) filterable by subscription plan, status (active/suspended/trial), signup date.

Business Rules:

- Suspending a Store (Platform Admin action, e.g., for Terms of Service violation or non-payment) immediately unpublishes its Public Storefront and blocks new Orders, but preserves all data (no deletion) pending resolution.
- Staff Permission Profiles (department-based permission templates referenced in §4.1) are defined at Store level and applied to Employees upon invite; changing a profile's permissions retroactively applies to all staff currently assigned to it.
- Tax rules support multiple rates (e.g., by product category or region) with a defined precedence order when multiple rules could apply to the same line item.

Validation Rules: Tax rate: 0–100%, numeric. Shipping rule: must define at least a flat rate or a weight/price-based tier before activation.

Permissions: Fully governed by §4 (Roles & Permission Matrix).

Edge Cases: A Store is suspended while it has pending Payouts: payouts are held (not cancelled) until suspension is resolved, to avoid disputes over merchant funds. A Staff Permission Profile is deleted while staff are still assigned to it: deletion is blocked until those staff are reassigned to another profile.

Notifications: Store suspended/reinstated (merchant), permission profile changed (affected staff, optional), tax/shipping rule changed (internal confirmation).

Audit Logs: All settings changes and every Platform Admin action are logged in `PlatformAuditLog`, which is itself immutable and independently retained even from the acting admin's own deletion permissions.

Acceptance Criteria: - [ ] Store suspension unpublishes the storefront and blocks new orders without deleting data. - [ ] Permission Profile changes propagate correctly to all currently-assigned staff. - [ ] All Platform Admin actions are captured in an immutable audit log.

Future Improvements: Self-service staff onboarding checklists; configurable approval workflows (e.g., discounts above X require Owner sign-off); Enterprise-tier custom roles beyond the four base roles.

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7. AI Features

AI is a first-class capability woven throughout FashionFlow AI, not a separate add-on. This chapter documents every AI feature in detail.

7.1 AI Product Writer

Business Value: Eliminates the time cost of writing product titles, descriptions, SEO metadata, and tags — work merchants otherwise do manually or outsource.

User Workflow: While creating/editing a Product, the merchant clicks "Generate with AI." The system uses the product's uploaded images, selected category, and any manually entered attributes (color, material, fit) as input. The merchant reviews the generated draft and can accept, edit, or regenerate.

Inputs: Product image(s), category, brand, any manually entered attributes.

Outputs: Product title, long-form description, SEO keyword list, product tags, meta description.

Example Prompt (internal, constructed from merchant inputs): "Generate a product title, description, SEO keywords, and tags for a women's oversized cotton t-shirt in black, category: Tops, brand: [Brand Name]."

Example Response (illustrative): - Title: "Oversized Cotton Crew Tee — Black" - Description: A short, benefit-oriented paragraph describing fit, fabric, and styling versatility. - SEO Keywords: oversized tee, black cotton t-shirt, women's crewneck - Tags: casual, cotton, oversized, everyday

Limitations: Generated copy may occasionally misidentify fine details not visible in the photo (e.g., exact fabric blend) — the system labels AI-generated fields clearly and does not auto-publish without merchant review.

Fallback Behavior: If AI generation fails (timeout, service error), the Product form falls back to standard manual entry with no data loss to already-entered fields; the merchant is notified generation failed and can retry.

Acceptance Criteria: - [ ] Generation completes within an acceptable time window (see §13 performance) or fails gracefully with a clear retry option. - [ ] Generated content is never auto-published without merchant review/acceptance. - [ ] All AI-generated fields are visually distinguishable from manually entered ones until accepted.

7.2 AI Fashion Designer

Business Value: Gives merchants access to design capability (prints, patterns, seasonal concepts) without hiring a designer, accelerating new-collection development.

User Workflow: The merchant provides either an existing clothing image, a blank template, or a text prompt describing a desired style. The AI generates design concepts (prints/patterns/color palettes) which the merchant can iterate on or export for use with a manufacturer.

Inputs: Existing clothing image, blank template selection, or free-text prompt.

Outputs: Clothing print designs, pattern variations, color palette suggestions, seasonal collection concepts.

Example Prompts: "Generate modern streetwear designs." / "Generate summer floral dress concepts."

Example Response (illustrative): A set of 3–4 generated print/pattern concepts with an accompanying palette, tagged by style descriptor (e.g., "Streetwear — Bold Graphic," "Streetwear — Minimal Typography").

Limitations: Generated designs are concepts/inspiration, not production-ready manufacturing files; they are not guaranteed to be free of resemblance to existing copyrighted designs and merchants are responsible for verifying originality before commercial production. The system does not reproduce or closely imitate identifiable copyrighted characters, logos, or branded IP.

Fallback Behavior: If a prompt is rejected (e.g., requests imitation of a specific copyrighted brand/character), the system explains that it cannot replicate protected designs and suggests describing the desired style attributes instead (color, silhouette, motif).

Acceptance Criteria: - [ ] Merchant can generate design concepts from an image, template, or text prompt. - [ ] The system declines prompts requesting reproduction of protected/branded designs, with a clear explanation. - [ ] Generated concepts are downloadable/exportable for external use.

7.3 AI Campaign Generator

Business Value: Removes the need for a dedicated copywriter to produce ad copy and captions across channels.

User Workflow: From the Campaign Builder, the merchant selects the products/collection being promoted and the target channel(s); the AI drafts copy variants which the merchant can edit before scheduling.

Inputs: Product/collection selection, target channel (Facebook, Instagram, WhatsApp, Email), campaign goal (e.g., flash sale, new arrival, win-back), tone preference.

Outputs: Facebook ad copy, Instagram captions, WhatsApp campaign messages, email campaign copy, promotional message variants, product launch ideas.

Example Prompt: "Write an Instagram caption for a flash sale on our summer dress collection, playful tone."

Example Response (illustrative): A short caption emphasizing urgency and the collection's key selling points, with suggested hashtags, kept within the channel's best-practice length.

Limitations: Generated copy reflects general best practices, not the merchant's unique brand voice, until the merchant has used the feature enough for the system to learn stylistic preferences (see Future Improvements). Cannot guarantee ad-platform policy compliance; merchant is responsible for final review before publishing to third-party ad platforms.

Fallback Behavior: If generation fails, the Campaign Builder allows manual copy entry without blocking the rest of the campaign creation flow.

Acceptance Criteria: - [ ] Generates channel-appropriate copy respecting each channel's length/format conventions. - [ ] Merchant can regenerate, edit, or manually override any generated copy before send.

7.4 AI Store Assistant

Business Value: Acts as an on-demand business advisor, surfacing recommendations a merchant would otherwise need an analyst to produce.

User Workflow: Accessible as a conversational assistant from the main dashboard; the merchant asks a question in natural language or selects from suggested prompts, and receives a data-grounded answer drawing on live Analytics.

Inputs: Natural-language question; underlying Analytics data (sales, inventory, campaign, customer).

Outputs: Direct answers to questions such as: Which products should be promoted? Which products are underperforming? Which customers are likely to buy again? What campaign should I run this week? Which products need restocking?

Example Prompt: "Which products should I promote this week?"

Example Response (illustrative): A short recommendation naming 2–3 specific products with the underlying reasoning (e.g., strong recent sell-through, healthy margin, approaching seasonal relevance), plus a one-click "create campaign" action.

Limitations: Recommendations are only as good as the underlying data volume — new stores with limited sales history receive more generic, best-practice-based guidance until sufficient data accumulates. The assistant does not execute actions autonomously (e.g., it will not launch a campaign without explicit merchant confirmation).

Fallback Behavior: If a question falls outside the assistant's grounded data scope, it says so plainly rather than fabricating a specific figure, and offers to help with what it can answer.

Acceptance Criteria: - [ ] Every data-grounded answer is traceable to an underlying Analytics metric (no fabricated numbers). - [ ] The assistant never autonomously executes a business action (campaign launch, discount, refund) without explicit confirmation. - [ ] New stores with insufficient history receive a clearly-labeled "general guidance" response rather than a false-confidence data claim.

7.5 AI Customer Insights

Business Value: Surfaces which customers need attention (win-back) and which are most valuable, enabling targeted retention effort.

User Workflow: Insights appear directly on Customer profiles and the Customer Analytics dashboard as scores/predictions, refreshed on a schedule.

Inputs: Customer purchase history, recency/frequency/monetary patterns, engagement with past campaigns.

Outputs: Churn risk score, repeat-purchase probability, customer value tier, product demand signals tied to customer preferences, best campaign timing per customer/segment.

Example Output (illustrative): A customer profile displaying "Churn risk: High — no purchase in 75 days, previously a Frequent buyer" alongside a suggested win-back campaign.

Limitations: Predictions are probabilistic estimates, not guarantees, and are recalculated periodically rather than instantaneously; scores are less reliable for customers with very few historical orders.

Fallback Behavior: Customers with insufficient order history (fewer than 2 orders) show as "Insufficient data" rather than a potentially misleading score.

Acceptance Criteria: - [ ] Scores refresh on a defined schedule and are timestamped with "last calculated" info. - [ ] Customers with insufficient history are explicitly labeled rather than scored.

7.6 AI Inventory Forecasting

Business Value: Reduces stockouts and overstock by predicting future demand per variant.

User Workflow: Forecasts appear on the Inventory dashboard and Purchase Order creation screen as suggested reorder quantities and timing.

Inputs: Historical sales velocity per variant, seasonality patterns, current stock levels, lead time from suppliers (if recorded).

Outputs: Suggested reorder point, suggested reorder quantity, forecasted stockout date at current sell-through rate.

Limitations: Forecast accuracy depends on sufficient sales history (minimum recommended: 60 days of sales data per variant); new products default to category-level average forecasting until enough of their own history accumulates.

Fallback Behavior: Variants without sufficient history display "Not enough data yet" instead of a low-confidence forecast presented as fact.

Acceptance Criteria: - [ ] Forecasts are visibly tied to a confidence basis (data volume) and never presented as certain. - [ ] Suggested reorder quantities can be applied directly to a new Purchase Order with one action.

7.7 AI Recommendation Engine

Business Value: Powers cross-sell/upsell opportunities on the storefront and in campaigns without manual merchandising effort.

User Workflow: Operates largely in the background — surfacing "customers also bought" and "recommended for you" placements on the Public Storefront, and suggesting audience/product pairings within the Campaign Builder.

Inputs: Aggregate purchase co-occurrence data, browsing behavior (where tracked), product attribute similarity.

Outputs: Product recommendation sets for storefront placements and campaign targeting suggestions.

Limitations: Cold-start problem for new stores/products with no purchase history — falls back to category/attribute-based similarity rather than behavioral data until sufficient volume exists.

Fallback Behavior: Defaults to merchant-curated "Featured Products" placement if insufficient data exists for personalized recommendations.

Acceptance Criteria: - [ ] Recommendation placements always render something sensible even with zero behavioral history (via the attribute-based/featured fallback). - [ ] Recommendations update as new purchase data accumulates without requiring manual reconfiguration.

7.8 Cross-Cutting AI Principles

- All AI-generated content is clearly labeled as AI-generated within the interface until a human reviews/accepts it.
- No AI feature autonomously executes a business-impacting action (publishing, sending, charging, discounting) without explicit user confirmation.
- AI generation requests are processed asynchronously via background jobs (see Non-Functional Requirements §13 and the Background Jobs architecture) so the interface remains responsive during generation.
- AI features degrade gracefully: every feature has a defined fallback when the underlying model/service is unavailable or when data is insufficient, rather than blocking the user's workflow.

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8. Analytics

8.1 Sales Analytics

Metric	Definition	Chart Type	Filters
Revenue	Sum of Order totals (net of refunds) in period	Line chart, trend over time	Branch, Channel, Date range
Orders	Count of completed Orders	Bar chart	Branch, Channel, Date range
Profit	Revenue minus cost of goods sold (from Product cost field)	Line chart	Branch, Date range
Growth Rate	% change vs. prior comparable period	KPI card with delta indicator	Comparison period toggle

Export options: CSV, PDF summary. **Business insight surfaced:** best/worst performing day of week, peak sales hours (useful for staffing).

8.2 Product Analytics

- **Top-selling products:** Ranked table by units sold and revenue, filterable by Category/Collection.
- **Slow-moving products:** Products with sell-through below a configurable threshold over a trailing period — feeds directly into AI Store Assistant recommendations.
- **Inventory value:** Total cost-basis value of current stock on hand, by Branch and Category.

8.3 Customer Analytics

- **Lifetime value distribution:** Histogram of customer LTV, with segment overlay (New/Frequent/VIP/Inactive).
- **Retention rate:** % of customers who made a repeat purchase within a defined window.
- **Repeat purchase rate:** Average number of orders per returning customer.

8.4 Campaign Analytics

- **ROI:** Campaign-attributed revenue divided by campaign cost (ad spend + promo discount value given).
- **Conversion rate:** % of audience recipients who completed a purchase attributable to the campaign (via promo code usage or tracked link/landing page attribution).
- **Revenue attribution:** Dollar amount of sales tied to a specific campaign, using last-touch attribution within a defined attribution window (default 7 days).

8.5 Traffic Analytics (Storefront)

- **Visitors:** Unique visitor count to Storefront/Landing Pages.
- **Device breakdown:** Mobile vs. desktop vs. tablet share.
- **Traffic sources:** Direct, social (Meta), search, campaign-link referral.
- **Conversion funnels:** Storefront visit → Product view → Add to cart → Checkout started → Order completed, with drop-off rate at each stage.

8.6 Financial & Operational Analytics

- **Financial:** Wallet balance trend, fees paid, payout history, tax collected summary.
- **Operational:** Staff productivity (orders processed per staff member at POS), average fulfillment time (order placed → fulfilled).

8.7 Dashboard & Export Standards

All dashboards support: date-range filter (with comparison-period toggle), Branch filter (for multi-branch stores), and CSV/PDF export. Charts refresh on the cadence defined in §6.13 (real-time counters for today; hourly aggregation for trend charts).

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9. Notifications

9.1 Notification Catalog

Event	Recipient(s)	Channels	Priority	Preference-Configurable
New order placed	Store Owner, Manager, assigned Sales staff	In-app, Email, WhatsApp (real-time via Reverb)	High	Yes
Payment failed	Store Owner	In-app, Email	High	No (financial-critical)
Low stock alert	Store Owner, Manager, Warehouse staff	In-app, Email	Medium	Yes
Out of stock	Store Owner, Manager, Warehouse staff	In-app, Email	Medium	Yes
Purchase Order received	Warehouse staff, Manager	In-app	Low	Yes
New customer message	Assigned staff, Manager	In-app, Web Push (real-time)	High	Yes
Campaign sent confirmation	Marketing Manager, Store Owner	In-app, Email	Medium	Yes
Campaign performance summary	Marketing Manager, Store Owner	Email	Low	Yes
Refund processed	Customer, Store Owner	Email (customer), In-app (staff)	Medium	Partially (customer notice not disableable)
New device login	Account owner	Email	High	No (security-critical)
Password/2FA changed	Account owner	Email	High	No (security-critical)
Subscription renewal reminder	Store Owner	Email, In-app	Medium	Yes
Payout sent/failed	Store Owner	Email, In-app	High	No (financial-critical)
Store suspended/reinstated	Store Owner	Email, In-app	High	No
AI generation completed	Requesting user	In-app (real-time)	Low	Yes

9.2 Delivery Channels

- **In-app:** Persistent Notification Center, real-time push via Laravel Reverb for time-sensitive events.
- **Email:** Transactional emails for confirmations, alerts, and summaries.
- **WhatsApp:** Used for both customer-facing order updates and, where opted in, staff alerts.
- **Web Push:** Browser-level push for real-time staff alerts (new message, new order) when the dashboard tab is not active.

9.3 Priority & Read Status

Notifications carry a priority (High/Medium/Low) affecting default channel selection and visual treatment in the Notification Center (High priority is visually highlighted and does not auto-collapse). All notifications track Read/Unread state per recipient; marking read is a per-user action and does not affect other recipients' read status for the same event.

9.4 Preferences

Each user manages their own channel preferences per event type where "Preference-Configurable" is Yes (see table above). Security- and financial-critical events cannot be fully disabled, though channel selection (e.g., email vs. in-app only) may be limited by the system to guarantee at least one reliable channel remains active.

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10. Integrations

10.1 Google OAuth

Why it exists: Reduces signup friction for both merchants (Store Owner signup) and, where applicable, storefront customers wanting a fast checkout login. **Business workflow:** A user selects "Continue with Google," authorizes access, and FashionFlow AI creates or matches an account by verified email. If an existing password-based account shares that email, the system prompts to link accounts (see §6.1 edge cases) rather than creating a duplicate.

10.2 Facebook OAuth

Why it exists: Mirrors the Google OAuth workflow for users who prefer Facebook login, and additionally enables the Facebook/Instagram ad integration used by AI Campaign Generator and Meta Pixel tracking (same underlying Meta account).

10.3 WhatsApp

Why it exists: The dominant customer communication channel for many fashion retail markets. **Business workflow:** Order confirmations, shipping updates, and opted-in marketing messages are sent via WhatsApp; customer replies flow into the unified Messaging Center (§6.10). Requires phone verification (OTP) on the store's business number before activation.

10.4 Email

Why it exists: Universal fallback channel for transactional communication (receipts, password resets, security alerts) and a primary channel for marketing campaigns/newsletters. **Business workflow:** All transactional email is sent regardless of marketing opt-in status (distinguishing transactional vs. promotional consent, a compliance requirement); promotional email honors unsubscribe preferences.

10.5 Payment Gateway

Why it exists: Enables the store to accept real payments online and, where supported, via card-present POS. **Business workflow:** Store Owner connects a supported payment gateway during Store Setup; the platform never stores raw card data itself (delegated to the gateway/tokenization), only references transaction/token IDs. Successful/failed payment webhooks drive Order status transitions (§6.6).

10.6 AWS S3

Why it exists: Scalable, durable storage for product images, store logos, and private assets (e.g., exported reports). **Business workflow:** All merchant-uploaded media is stored in S3 with access governed by signed URLs — private assets (e.g., a generated financial export) are only accessible via time-limited signed links, never public by default.

10.7 Google Sheets

Why it exists: Many small fashion merchants already manage parts of their business in spreadsheets; this integration bridges that habit into the platform. **Business workflow:** Supports bulk product import/export and inventory reconciliation via a connected Google Sheet, reducing reliance on manual CSV upload/download for merchants already living in Sheets.

10.8 Meta Pixel

Why it exists: Enables merchants to measure and optimize Facebook/Instagram ad performance against actual Storefront conversions. **Business workflow:** Once connected, Storefront page views, add-to-cart, and purchase events are relayed to the merchant's Meta Ads account, closing the loop between ad spend and real sales — feeding the Campaign Analytics ROI calculation (§8.4).

10.9 Google Tag Manager

Why it exists: Gives more sophisticated merchants (or their agencies) flexibility to add additional tracking/tag configurations without engineering changes on the FashionFlow AI side. **Business workflow:** Store Owner pastes their GTM container ID into Storefront settings; all subsequent tag management happens within their own GTM account.

10.10 Microsoft Clarity

Why it exists: Gives merchants free session-replay and heatmap insight into how customers actually use their Storefront, informing landing page and merchandising decisions beyond what standard Analytics numbers show. **Business workflow:** Store Owner connects a Clarity project ID; behavioral insights are viewed in Clarity's own dashboard (external to FashionFlow AI) but referenced in onboarding/education content as a recommended complement to platform Analytics.

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11. Business Rules

This chapter consolidates all cross-cutting business rules referenced throughout the document into a single authoritative reference.

11.1 Product Availability

A Product is visible on the Storefront only if: status = Active, at least one Variant has stock > 0 OR backorder is explicitly enabled for that variant, and the parent Store's Storefront is published (see §6.9 completeness gate).

11.2 Payment Method Requirement

A Store cannot accept live Orders (Online or POS) until at least one Payment Method is configured and validated (§6.12).

11.3 Backorder / Pre-Order Exception

By default, stock cannot go negative (§6.4). Merchants may explicitly enable "Allow backorder" per Variant, which permits sales beyond zero stock; such Orders are flagged "Backordered" in the pipeline and do not auto-transition to "Fulfilled" until stock is replenished and staff confirm shipment.

11.4 Discount Stacking

Only one discount source applies per line item by default: the system compares (a) the Product's own discount price, (b) any applicable Promo Code, and (c) any active Campaign-level discount, and applies whichever yields the customer the greatest single benefit — unless a Promo Code or Campaign is explicitly flagged "stackable," in which case it may combine with a Product's base discount price (but never with another non-stackable Promo Code/Campaign simultaneously).

11.5 Refund Thresholds

Sales Employees may process refunds up to a configurable threshold (default: local-currency equivalent of $100) without approval. Refunds above the threshold require Manager or Store Owner sign-off before the refund is finalized; the request is held in a "Pending Approval" state visible to approvers.

11.6 Customer Segmentation

Segments (New/Frequent/VIP/Inactive) are recalculated nightly per the definitions in §6.7. Manual VIP tags override automatic reclassification until removed by staff.

11.7 Campaign Priority

When multiple active Campaigns could apply conflicting discounts to the same customer/product combination, the Campaign with the more specific targeting (e.g., a Flash Sale on a specific Collection) takes priority over a broader campaign (e.g., a store-wide Seasonal discount); if specificity is equal, the more recently created Campaign takes priority. The customer is never shown or charged for more than one discount on the same line item.

11.8 Inventory Adjustments

All stock changes must flow through a typed `InventoryTransaction` (§6.4); direct silent edits to stock quantity are not permitted anywhere in the system, including by Platform Admins.

11.9 Data Ownership & Multi-Branch Sharing

Each Branch's stock is independently tracked by default. A Store Owner may enable "Shared Inventory Pool" mode for a multi-branch Store, in which case Variant stock is tracked once at the Store level and all Branches deduct from the same pool — this setting cannot be toggled back to per-branch tracking without a manual reconciliation step (to avoid ambiguous stock-splitting). Customer data (CRM) is shared across all Branches of a single Store Owner by default, since a customer typically shops across a brand's branches regardless of location.

11.10 Archiving vs. Deletion (Soft Delete)

Products, Campaigns, and Customers are never hard-deleted while referenced by historical Orders/transactions; "deletion" instead sets an archived/inactive flag, hiding the record from active lists and the Storefront while preserving referential integrity for financial and audit records.

11.11 Account & Data Deletion Rules

- A Store Owner account cannot be deleted while it owns active (non-transferred, non-deactivated) Stores.
- Customer data-deletion requests anonymize the Customer record (name/contact info scrubbed) while preserving the associated Order records' financial figures for tax/accounting retention requirements.
- Deleted records enter a grace period (default 30 days) during which a Store Owner can request restoration before permanent purge, except where a customer has explicitly exercised a legal deletion right.

11.12 Concurrency

Stock-affecting operations (Orders, Adjustments, Purchase Order receiving) use row-level locking/optimistic concurrency control on the affected `StockLevel` record to prevent overselling under simultaneous requests (e.g., two POS terminals or an online order racing a POS sale for the last unit). The losing request in a race is rejected with a clear "no longer available" message rather than allowed to silently succeed into negative stock.

11.13 Validation & Conflict Resolution (General Principles)

- Server-side validation is authoritative in all cases; client-side validation is a UX convenience only and never the sole enforcement point.
- Where two staff members edit the same non-stock record concurrently (e.g., a Product description), a "last write wins" model applies but the losing editor is shown a warning if their view was stale at save time, with an option to review the current version before overwriting.

---

12. User Flows

12.1 Customer Purchase (Online)

```
Customer browses Storefront
   → Views Product → Selects Variant (size/color)
   → Adds to Cart
   → Proceeds to Checkout
   → Enters/confirms shipping address
   → Selects payment method → Payment authorized
   → System re-validates stock at payment capture (§11.12)
   → Order created (status: Processing) → Stock decremented
   → Confirmation shown to customer + notification sent (§9.1)
   → Order appears in Order Management for fulfillment
```

12.2 Inventory Update (Purchase Order Receiving)

```
Manager/Warehouse creates Purchase Order (Draft)
   → Adds line items (Variant + quantity + cost) → Sends to Supplier (status: Sent)
   → Shipment arrives → Warehouse opens Receiving screen
   → Records actual received quantity per line item
   → If full quantity received → PO status: Received
   → If partial → PO status: Partially Received (remainder stays open)
   → Stock levels increased via InventoryTransaction (type: purchase_order_receipt)
   → Notification sent to Manager/Owner
```

12.3 Campaign Creation

```
Marketing Manager opens Campaign Builder
   → Selects Campaign Type (Product/Seasonal/Flash Sale/Discount)
   → Selects Audience (segment or custom filter) → System validates non-zero recipient count
   → (Optional) Uses AI Campaign Generator to draft copy per channel
   → Sets discount/promo code (if applicable) → System checks stacking rules (§11.4)
   → Sets Schedule (start/end) and Budget
   → Reviews summary → Submits for Store Owner approval if required (§4.3/§6.8)
   → Campaign status: Scheduled → Sent at scheduled time → Status: Active/Completed
   → Performance tracked in Campaign Analytics (§8.4)
```

12.4 AI Design Generation

```
Merchant opens AI Fashion Designer
   → Provides input: existing image, blank template, or text prompt
   → Submits generation request → Job queued (background, §Background Jobs)
   → Real-time progress notification (Reverb) while processing
   → Generated concepts returned → Merchant reviews set of design variations
   → Merchant accepts, downloads, or requests regeneration
   → Accepted design optionally attached to a Product or Collection for reference
```

12.5 Product Publishing

```
Merchant creates Product (Draft) → Adds Category/Brand/Collection/Tags
   → Adds Variants (size/color/SKU/price/cost)
   → Uploads images (at least 1 required)
   → (Optional) Uses AI Product Writer for title/description/SEO
   → System validates: price set, SKU unique, image present (§6.3 validation)
   → Merchant sets status: Active
   → Product becomes visible on Storefront (subject to §11.1 availability rule)
```

12.6 Order Fulfillment

```
Order status: Processing (payment confirmed)
   → Warehouse/Sales staff view Order in fulfillment queue
   → Item picked/packed (online) or handed over (POS, immediate)
   → Staff marks status: Fulfilled
   → Stock already decremented at Processing stage — no further inventory action
   → Customer notified (online) / receipt provided (POS)
   → Order timeline updated (§6.6 audit logs)
```

12.7 Refund

```
Staff opens Order Detail → Selects "Process Refund"
   → Enters refund amount (≤ remaining refundable balance, §6.6 validation)
   → If amount ≤ staff's threshold (§11.5) → Refund processed immediately
   → If amount > threshold → Refund enters "Pending Approval" → Manager/Owner notified
   → Approver approves/rejects → If approved, Refund processed
   → Refund triggers Payment reversal + InventoryTransaction (type: return) unless marked non-restockable
   → Customer notified → Order status updated to Refunded (full) or remains Fulfilled with partial refund no
```

12.8 Customer Support (Messaging)

```
Customer sends message via WhatsApp or Email
   → Message routed into Messaging Center, matched to existing Customer profile or creates new Conversation
   → Assigned staff (or any staff with access) views thread with full order/purchase context
   → Staff replies (manually or via Quick Reply template)
   → Staff adds Internal Note if escalation/context needed for a colleague
   → Conversation marked Resolved when concluded → Reopens automatically if customer replies again
```

13. Non-Functional Requirements

13.1 Performance

Area	Target
Storefront page load (P95)	Under 2.5 seconds on standard broadband
Dashboard/report load (P95)	Under 3 seconds using pre-aggregated snapshots
POS transaction completion	Under 1 second server response for stock check + order creation
AI generation (Product Writer/Campaign copy)	Under 15 seconds typical; asynchronous with progress indicator beyond that
AI Fashion Designer generation	Under 60 seconds typical; asynchronous with progress indicator
Real-time notification delivery (Reverb)	Under 2 seconds from event to client display

13.2 Availability

Target platform uptime of 99.9% monthly, excluding scheduled maintenance windows (communicated at least 48 hours in advance for anything customer-facing-impacting). Core transactional paths (Checkout, POS, Payment processing) are prioritized above secondary features (AI generation, Analytics) in incident-response severity classification.

13.3 Scalability

The platform must support multi-tenant scaling where a single large merchant's load (e.g., a flash-sale traffic spike) does not degrade performance for other tenants (queue-based isolation for heavy operations — AI generation, bulk imports, campaign sends — via the Background Jobs architecture, monitored through Horizon).

13.4 Security

- All data in transit is encrypted (TLS).
- Payment data is never stored directly by the platform; only gateway tokens/references are retained (§6.12).
- Role-based access control (§4) is enforced at the API/service layer, not merely hidden in the UI.
- Admin impersonation is time-boxed and fully audit-logged (§4.5).
- Regular dependency and vulnerability scanning as part of CI/CD.

13.5 Accessibility

Customer-facing Storefront and core merchant dashboard screens target WCAG 2.1 AA conformance: sufficient color contrast, keyboard navigability, screen-reader-compatible labeling for interactive elements, and accessible form error messaging.

13.6 Localization

- English and Arabic supported at launch, with full RTL layout support for Arabic.
- All user-facing strings are externalized to translation files (no hardcoded copy) to support adding further languages without structural rework.
- Currency, date, and number formatting respect the Store's configured locale/currency settings (§6.2).

13.7 Auditability

Every module in Chapter 6 defines its own audit-log requirements; at a platform level, audit logs are immutable, timestamped, attributed to an acting user or system process, and retained for a minimum compliance period (default: 24 months, configurable for Enterprise tier).

13.8 Monitoring & Logging

- Application performance and error monitoring via Sentry.
- Queue/job health monitored via Laravel Horizon.
- System-level performance metrics (request latency, memory, DB query time) via Laravel Pulse.
- Structured application logs support correlation IDs across a single request/job chain for effective debugging.

13.9 Disaster Recovery

- Automated, regularly tested database backups with a defined Recovery Point Objective (RPO) of no more than 24 hours (target: much lower for transactional data via continuous replication where feasible).
- Defined Recovery Time Objective (RTO) for full platform restoration in a disaster scenario, communicated in the platform's operational runbook (implementation detail owned by Engineering/Ops, referenced here as a product requirement that such a plan must exist and be tested periodically).

---

14. Future Roadmap

14.1 Version 2

- Loyalty points and rewards program tied to Customer segmentation.
- A/B testing for Campaigns and Landing Pages.
- Supplier portal for direct purchase-order confirmation.
- Digest-mode notifications (daily/weekly email summaries).

14.2 Version 3

- Buy-online-pickup-in-branch (BOPIS) fulfillment flow.
- Loyalty-linked installment/layaway payment plans.
- Natural-language querying of Analytics via the AI Store Assistant ("ask anything" reporting).
- Full visual theme customization for the Storefront (custom CSS/component-level control).

14.3 Enterprise Edition

- Custom roles beyond the four base roles, with granular permission-building tools.
- Configurable multi-step approval workflows (e.g., discounts above a threshold require Owner sign-off).
- Extended audit-log retention and dedicated compliance export tooling.
- SSO (SAML/OIDC) for large multi-brand organizations.

14.4 AI Roadmap

- Brand-voice learning for AI Campaign Generator and AI Product Writer (adapts to a merchant's historical accepted edits over time).
- Automated, AI-suggested reorder execution (with merchant approval-in-the-loop) building on AI Inventory Forecasting.
- AI-assisted customer service reply drafting within the Messaging Center.
- Predictive segment-transition modeling (forecasting which customers are about to become Inactive or VIP, ahead of the nightly recalculation crossing that threshold).

14.5 Mobile Application

- Native mobile app for Store Owners/Managers (dashboard, notifications, approvals on the go).
- Native mobile POS for Sales Employees on tablet/handheld hardware.
- Push notification parity with the current Web Push/Email/WhatsApp channel set.

14.6 Marketplace

- A multi-merchant marketplace surface allowing customers to browse across participating FashionFlow AI stores in one discovery experience (opt-in for merchants).
- Cross-store promotional placements and marketplace-level analytics.

14.7 API Platform

- Public REST API for merchants/partners to integrate FashionFlow AI data into external systems (accounting, ERP, custom storefronts).
- Webhook system for real-time event delivery (order created, stock changed, campaign sent) to external endpoints.
- Developer documentation and sandbox/testing environment.

End of Document — FashionFlow AI PRD v1.0